

TECHNICAL SKILLS

Proficient Languages: Python, R, SQL

Tools & Libraries: Jupyter Notebook (Anaconda), Sklearn, ggplot, Excel, Git, LaTeX

Machine Learning & Stats: KNN, K-Means Clustering, PCA, GLMs, EM-Algorithm

EDUCATION

Master of Mathematics and Statistics (MMathStat): 1st Class (75% so far), University of Warwick July 2027

Relevant Modules: Maths of Machine Learning, Machine Learning Frameworks, Multivariate Statistics, Generalised Linear Models, Python for Data-Analytic Tasks, Applied Statistical Modelling, Probability Theory

Mathematics A*, Further Mathematics A*, Economics A, Notre Dame High School - Sheffield Sep 2023

PROJECTS

2026 FIFA World Cup Monte Carlo Simulator | Python / Streamlit

- Built a Hierarchical Bayesian Negative Binomial model in PyMC to estimate latent attacking and defending strengths across 284 teams, applying MCMC sampling and Laplace-smoothed penalty priors.
- Simulated 1,000 World Cup brackets (104,000+ matches) to compute knockout survival probabilities, deploying an interactive Streamlit dashboard to analyse match overdispersion and extreme tail events.

Premier League Stratification Analysis | R / Excel / LaTeX

- Engineered a machine learning pipeline to challenge the "Big 6" narrative by extracting and analysing multi-dimensional team performance metrics.
- Applied Principal Component Analysis (PCA) for dimensionality reduction and K-Means clustering to identify hidden similarities and group teams into distinct statistical profiles.

Car Price Prediction Engine | R / LaTeX

- Conducted Exploratory Data Analysis (EDA) on a real-world Kaggle dataset, utilising K-Nearest Neighbours (KNN) to robustly impute missing data points.
- Developed and tuned Ridge, Lasso, and Stepwise regression models to predict car prices, evaluating accuracy and preventing overfitting using R^2 and Adjusted R^2 metrics.

EXPERIENCE

Barnett Waddingham

Actuarial Consulting Intern

Leeds, UK

July – Aug 2025

- Supported actuarial valuations for pension schemes with 5,000+ members by validating 10+ cash equivalent transfer values and reconciling member data using Excel.
- Produced and updated multiple client actuarial reports and accounting disclosures, ensuring accuracy and delivering work to tight deadlines.
- Drafted professional client communications and presented funding valuations to mock trustees.

Sheffield Scitt

Maths Teacher Intern

Sheffield, UK

Jun – Jul 2024

- Applied diverse teaching techniques to maintain engagement across varied ability groups.
- Led tutor groups of 5–10 students, supporting a 50% improvement in student test scores.

INTERESTS

- Active member of the University Statistics Football team (automated an Excel fantasy league data pipeline), keen squash player.
- Grade 5 pianist focusing on learning cinematic film scores, enthusiastic home cook who enjoys trying new recipes.